

10 RAG accuracy techniques

Your RAG is *lying to you.*

10 battle-tested techniques to make it actually accurate.

Hallucinations → **Accuracy ✓**



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Basic RAG isn't enough.

01

It hallucinates

Confidently states facts that don't exist.

02

It misses the answer

Skips the one document that actually had it.

03

It's inconsistent

Different answers to the same question.

04

It breaks in prod

Accuracy drops the moment real users arrive.

LLMs have a knowledge cutoff. RAG is the fix — only if you build it right.



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Don't optimize blindly.

The winning formula is a loop, not a guess:

1

Baseline

Stand up a naive RAG pipeline first.

2

Measure

Score it with a framework like RAGAS.

3

Fix one weak link

Attack a single bottleneck at a time.

4

Iterate

Re-measure, then repeat the loop.

The 10 techniques that move the needle →



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01 Hybrid Search

Vectors alone miss the obvious

- ✗ Pure vector search misses exact terms — codes, names, part numbers
- ✗ Pure keyword search misses meaning
- ✓ Hybrid = BM25 + vector similarity

Impact 3–3.5× recall • +11–15% answer accuracy

Best for — Legal docs, technical manuals, medical records



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02 **Tune Your HNSW Index**

Defaults leave accuracy on the table

- M — connections per node (denser = better recall)
- EF_CONSTRUCTION — build-time search depth
- EF_RUNTIME — query-time search depth
- ✗ Most teams ship the defaults. Most teams shouldn't.

Best for — Near-duplicate docs, FAQ variants, product specs



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03 Smarter Chunking

Stop splitting mid-thought

- ✗ Fixed-length chunks break ideas in half
- ✓ Split on paragraph breaks
- ✓ Split on section headers
- ✓ Split on semantic boundaries (topic shifts)

Impact Each chunk = one complete thought

Best for — Policy manuals, medical records, structured docs



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04 Fine-tune Embeddings

Generic models don't speak your language

- “Cardiac arrhythmia” $\approx$ “irregular heartbeat” — to a doctor
- ✗ To a generic model? Not so much
- ✓ Fine-tune embeddings on domain-specific pairs

Impact Small fine-tuned models beat huge generic ones

Best for — Law, finance, pharma, healthcare



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05 **Fine-tune the LLM**

Retrieval is right. Response is wrong.

- ✗ Cites the docs weirdly
- ✗ Wrong tone, skips compliance language
- ✓ Use LoRA to fix behavior — no full retrain

Best for — Healthcare, finance, regulated industries



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06

Semantic Caching

Same question → same answer, every time

- “Reset my password” = “I forgot my login” — same intent
- ✗ No cache → 2 calls, 2 different answers
- ✓ Semantic cache → 1 consistent response in ms

Impact Up to 73% cost reduction in chat workloads

Best for — FAQs, product docs, stable knowledge bases



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07 Long-term Memory

Stop making users repeat themselves

- ✗ Dumping full chat history → lost-in-the-middle
- ✓ Semantic memory → surface only relevant past turns
- Lean context = better reasoning

Best for — Coaching apps, support agents, personalized assistants



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08 Query Transforms

Users don't talk like documents

- User: "What's broken with my internet?"
- Docs: "Network connectivity diagnostics"
- ✓ HyDE — generate a fake answer, use it as the query
- ✓ Multi-query — rewrite 3–5 ways, retrieve in parallel

Best for — Vague queries, open-domain systems



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09 LLM as Judge

Who's grading your RAG?

- ✓ Faithfulness — is the answer grounded?
- ✓ Relevance — does it address the question?
- ✓ Context precision — were the right docs pulled?

Impact >80% agreement with human raters

Best for — High-stakes apps — legal, healthcare, enterprise



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10 Re-ranking

Order matters more than you think

- LLMs attend most to context that appears early
- ✗ Best doc ranked #8 → accuracy lost
- ✓ Cross-encoder scores query + doc together (top 20–100)

Impact 10–40% precision improvement

Best for — Noisy corpora, multi-source retrieval



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One fix per weak link.

Problem

Use this

Missing exact terms

→ **Hybrid Search**

Slow / weak retrieval

→ **Tune HNSW**

Fragmented context

→ **Smart Chunking**

Domain language gaps

→ **Fine-tune Embeddings**

Wrong tone / format

→ **Fine-tune LLM**

Inconsistent answers

→ **Semantic Caching**

No chat memory

→ **Long-term Memory**

Vague user queries

→ **Query Transforms**

Can't measure quality

→ **LLM as Judge**

Right docs, wrong order

→ **Re-ranking**

Pick your weak link. Fix it. Repeat.



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Building production RAG?

***Ship the
accurate one.***

Comment **“RAG”** and I’ll send you the full accuracy checklist.

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